RASST: Fast Cross-modal Retrieval-Augmented Simultaneous Speech Translation

Jiaxuan Luo, Siqi Ouyang, Lei Li

CMU

Outline

- Motivation
- Method
- Experiment Setup
- Results & Analysis

Motivation

Simultaneous Speech Translation (SST)

- Generating translations incrementally as partial speech input is received.

- Widely used in international conferences, cross-lingual conversations, and etc.

Challenges of SST

- The speech can be very long, lasting hours
- Need to balance the translation quality and the latency
- **Domain-specific terminologies are hard to recognize and translate**

Example (Recognition Error)

Source Speech: More recent work has used masked language models to fill in masked portions of the text to change labels.
Reference: 最近的工作使用掩码语言模型来填充文本的掩码部分以更改标签。
Plain SST: 近期的研究则使用大规模语言模型来填补文本中被遮蔽的部分，以改变标签。
RASST: 近期的研究使用了掩码语言模型来填充文本中被掩码的部分，以改变标签。

Scenario: Source speech mentions "masked language models".
Failure Case (Plain SST): Translates to "大规模语言模型" (Mass Language Models)
Success Case (RASST): Correctly translates to "掩码语言模型" (Masked Language Models) by retrieving the correct term.

Example (Translation Error)

Source Speech: I am very glad to present our work on Online Semantic Parsing for Latency Reduction in task-oriented dialogue.
Reference: 我很高兴向大家介绍我们在在线语义解析用于减少面向任务的对话中的延迟方面的工作。
Plain SST: 非常高兴向大家介绍我们关于任务导向对话中语义解析延迟减少的工作。
RASST: 我非常高兴能介绍我们关于面向任务的对话中延迟降低的在线语...

Scenario: Source mentions the term “task-oriented dialogue”.

Failure Case (Plain SST): Translates it as “任务导向对话” (wrong term variant vs glossary/reference).

Success Case (RASST): Uses the correct term “面向任务的对话” as in the reference/glossary.

Retrieval-Augmented SST?

- Professional simultaneous interpreters use glossaries to help translating terminologies correctly

- Like prior ASR and MT research, we could introduce a retriever to retrieve relevant terminologies that could appear in speech from a given glossary
  - Problem solved?

Retrieval-Augmented SST? Unique Challenges

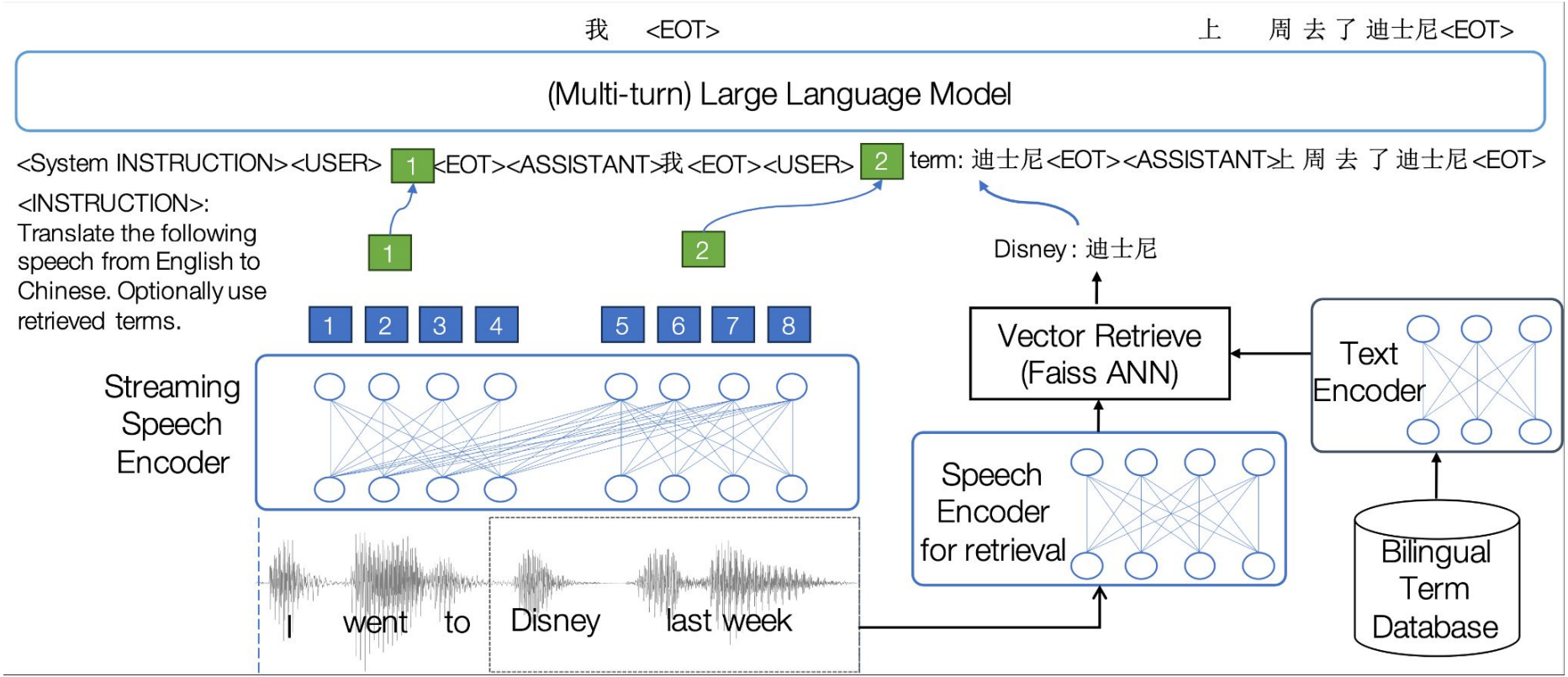
- **Cross-Modal Gap:** Input is streaming audio, glossary is text. Need to bridge the modality gap.

- **Latency Constraints:** Retrieval must happen in real-time on streaming input. No time for slow searches.

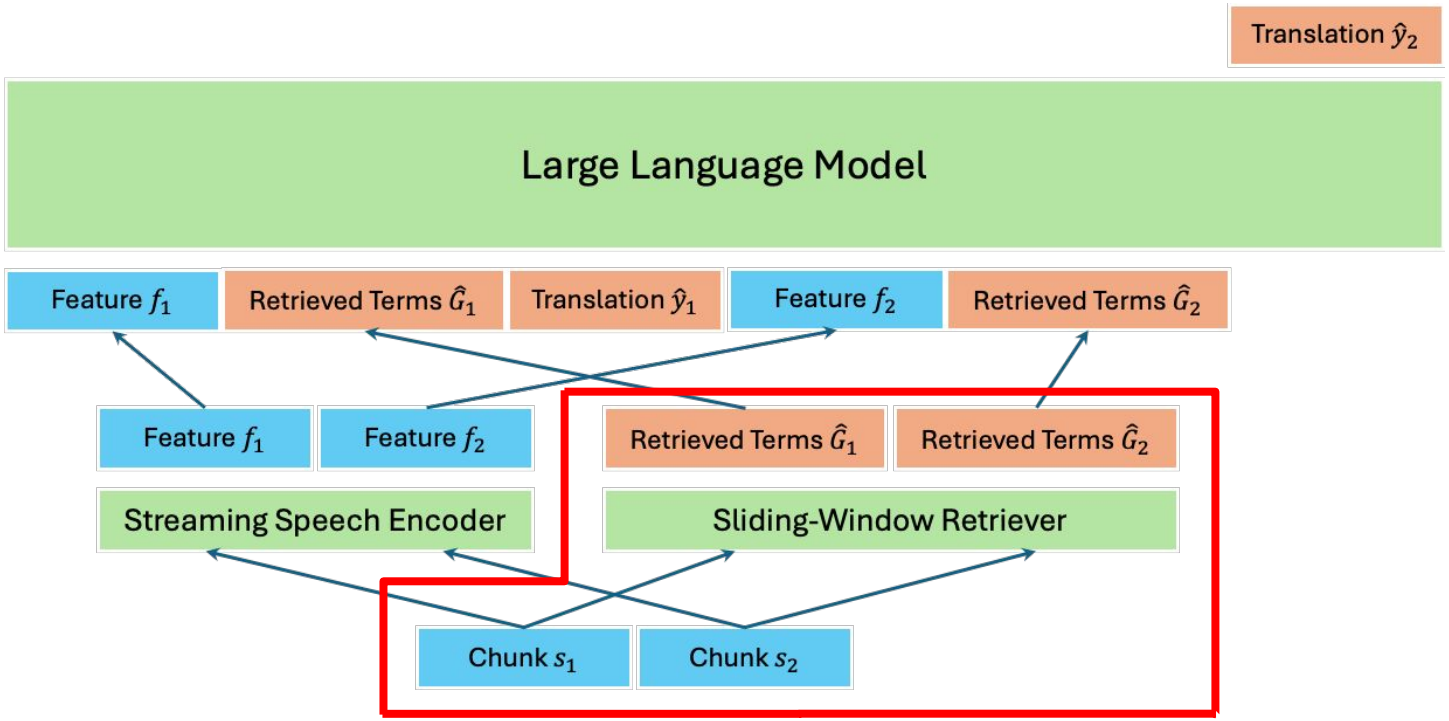
- **The "When" Problem:** Translation lags behind speech. The model must decide *if* and *when* to use a retrieved term.

Method: RASST

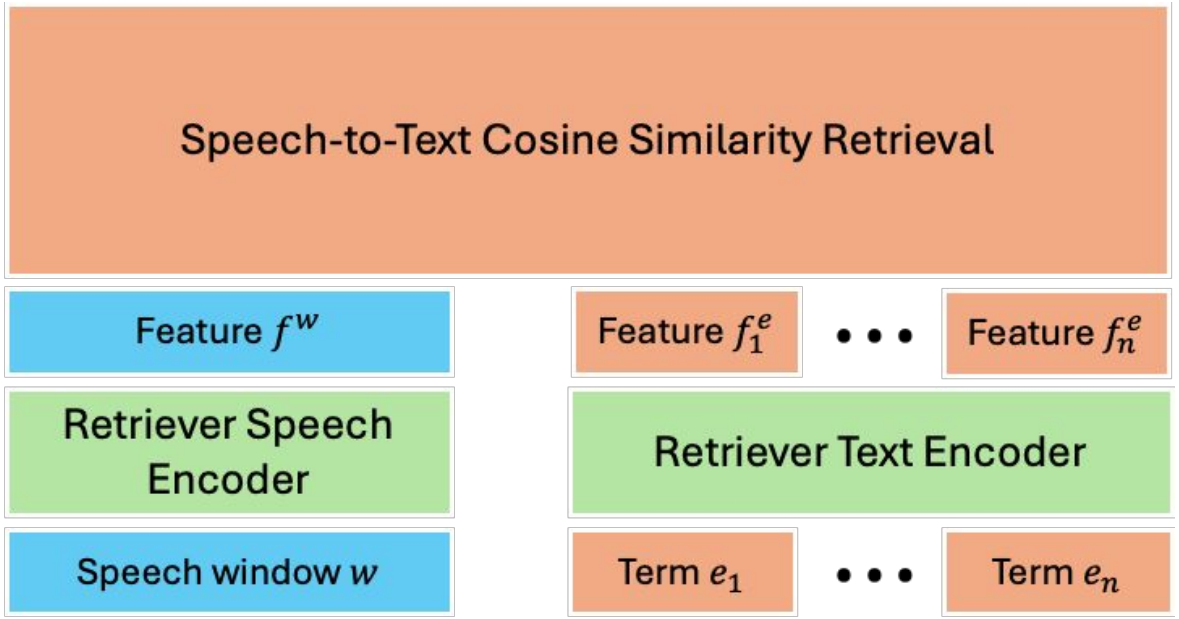
RASST Overview



RASST Overview



Retriever Architecture



Retriever Training

Motivation: Why Nouns/Phrases as "Terms"?

- High Information Density
- Form Stability
- Consistency

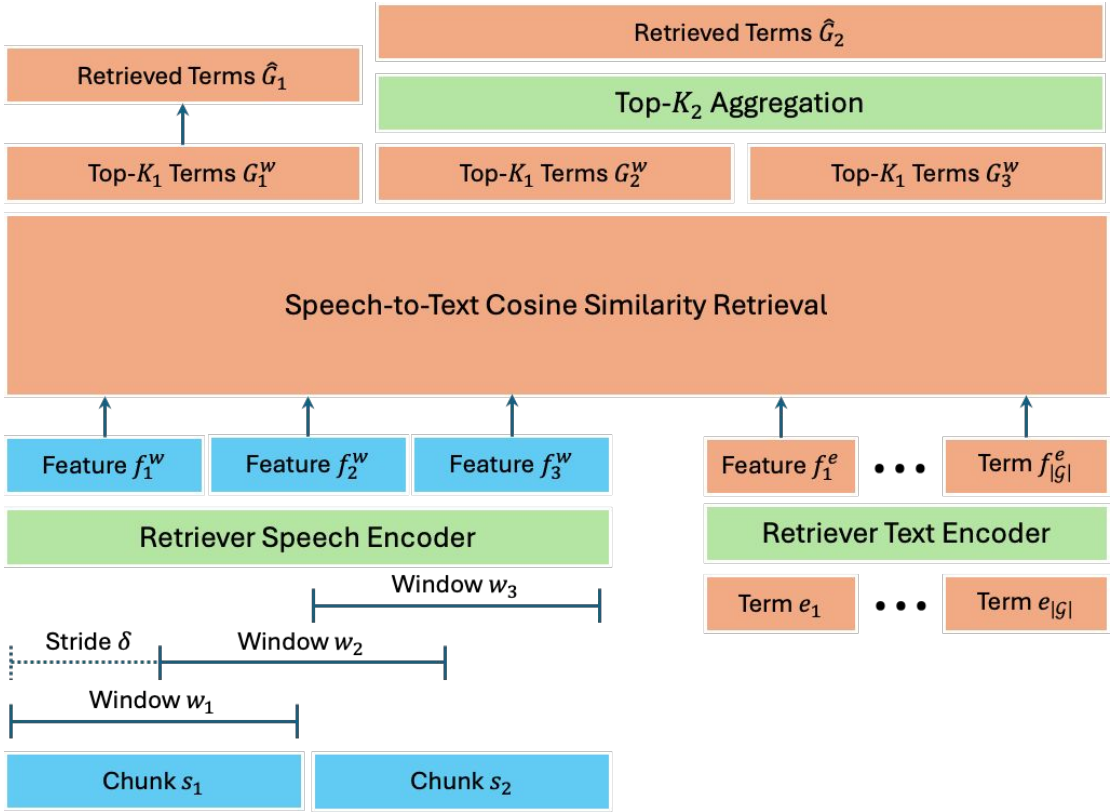
Learning Objective: Symmetric InfoNCE

- Contrastive Loss
- Negative Sampling: In-batch Negatives
- Multi-Positive Support

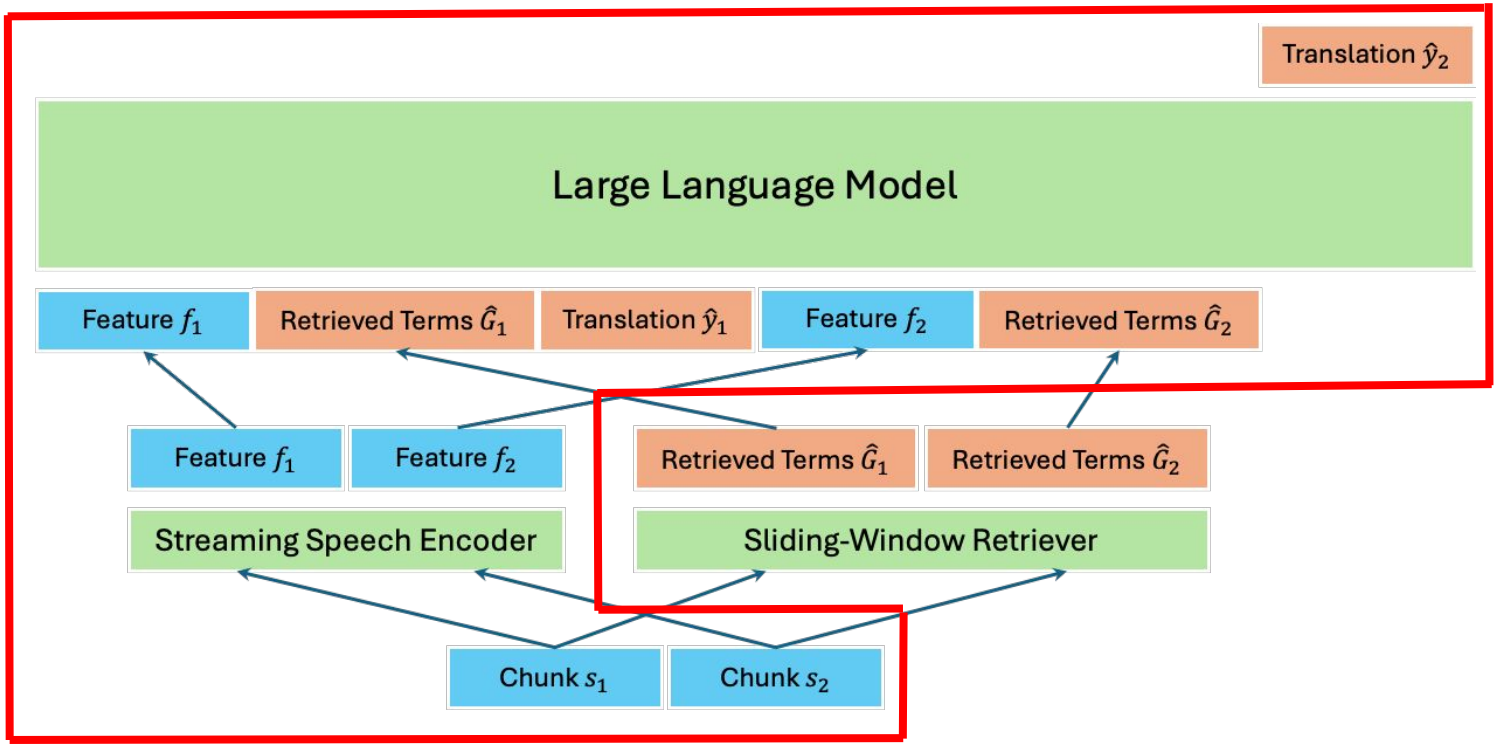
$$-\mathop{\mathbb{E}}_{s,\boldsymbol{t}^{\text{pos}},\boldsymbol{t}^{\text{neg}}}\left[\log\frac{\sum_{t\in\boldsymbol{t}^{\text{pos}}}\exp\left(\text{sim}(f_s,f_t)/\tau\right)}{\sum_{t\in\boldsymbol{t}^{\text{pos}}\cup\boldsymbol{t}^{\text{neg}}}\exp\left(\text{sim}(f_s,f_t)/\tau\right)}\right]$$

Role	Data Content
Anchor (Audio)	`.../term_dev_audio_chunks/POD1000000004_85_chunk_2.wav`
Context (ASR)	*"particular brand of **utopian fiction**"*
Context (Translation)	*"特定类型的 乌托邦小说"*
Target (Positive)	✅ **"utopian fiction"** (Glossary Term)
Negatives (In-Batch)	❌ "miracle" *(from chunk_5)* ❌ "vegetables" *(from chunk_6)* ❌ "brand" *(from chunk_1)* *... (other terms in batch)*

Inference: Sliding-Window Retrieval



RASST Overview



Training Speech LLM

- Prompt Structure:

  [System] *Simultaneous interpreter: translate English audio to Chinese. Use term_map if provided.*

  [User] <audio> + term_map: term=translation ... *(retrieved candidates)*

  [Assistant] outputs incremental translation for this chunk *(can be empty = "wait")*

- We need to simulate retrieved candidates

- Translation is obtained through word alignment from a prior work

Training Speech LLM - Data Synthesis

- **Method:** We construct training samples with three types of retrieved contexts
  - Standard: Ground-truth terms + Hard Negatives
    - Simulating imperfect retrieval
  - None: Empty set
    - Model must rely on audio only
  - All-Wrong: Only incorrect terms
    - Model must learn to ignore bad retrieval

- **Goal:** robust against retriever errors

Training Speech LLM - Hard Negative Mining

- **Observation:** Random negatives are too easy.

- **Strategy:** Use retriever text encoder to find "Hard Negatives", i.e., terms that are semantically similar but wrong
  - **Idea:** mine *similar-but-wrong* terms via **text encoder**
  - **Offline:** encode all terms → build **FAISS**
  - **Per GT term:** retrieve **Top-K** neighbors → remove GT / self
  - **Pick:** sample from **Top-N** most similar
  - **Final:** cap **20** terms in term_map

Experiment Setup

Datasets

- **Training:** GigaSpeech
  - Retriever training: 4M audio-term pairs
  - Speech LLM training: 12.5K instances with synthetic terms and translations

- **Evaluation:** ACL 60/60 Dev Set (academic talks, highly technical)
  - Tagged Glossary
  - Paper-Extracted Glossary **(**extracted from PDFs using Gemini-2.5-Pro)

Model Configuration

- Retriever
  - Speech encoder: Qwen3-Omni Audio Transformer
  - Text Encoder: BGE-M3

- Speech LLM
  - Qwen3-Omni-30B-A3B-Instruct

Metrics

- Translation Quality
  - BLEU
  - Terminology Translation Accuracy

- Latency: StreamLAAL

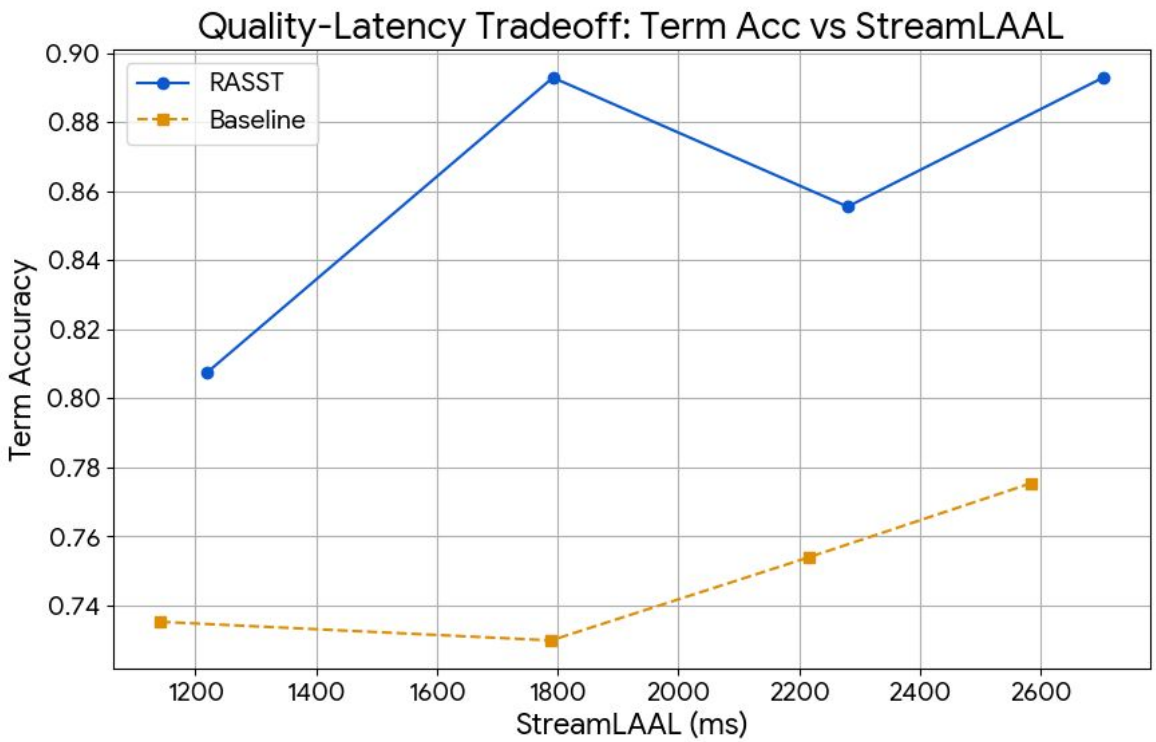
Baselines

- Offline ST with Qwen3-Omni-30B-A3B-Instruct
  - Serve as upper bound for translation quality

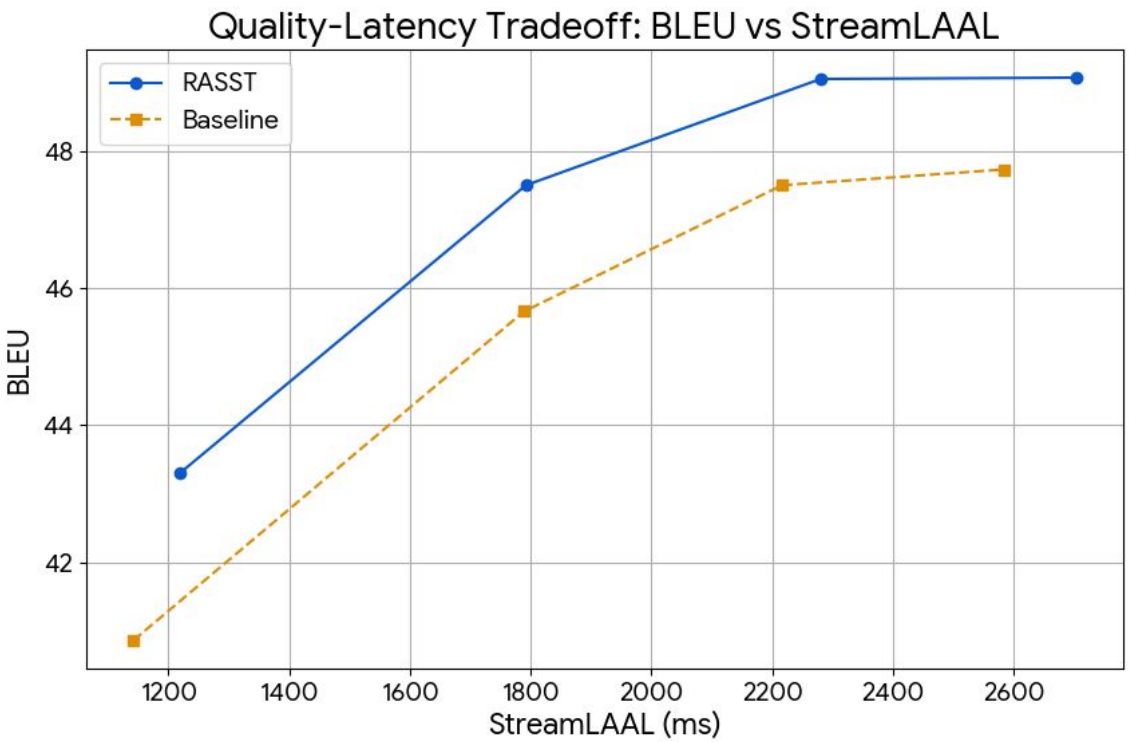
- InfiniSST: SOTA SST model
  - Equivalent to RASST without retrieval

Results & Analysis

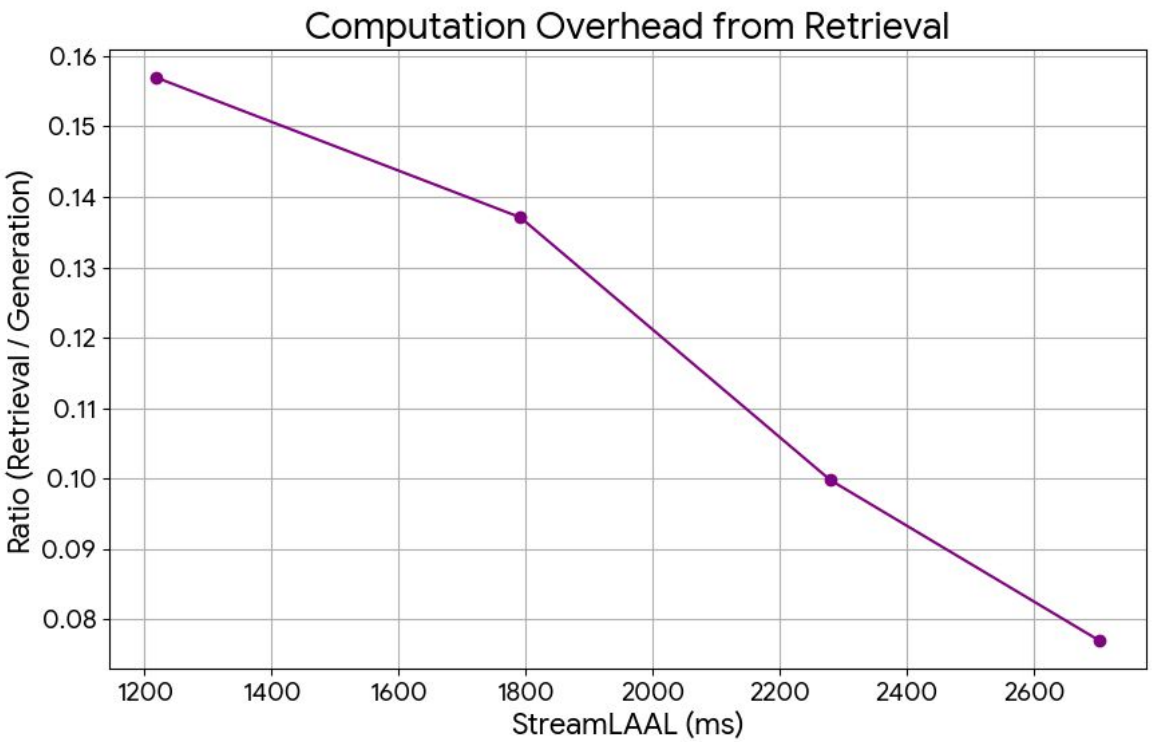
RASST Improves Terminology Translation Accuracy



RASST Improves Translation Quality



RASST Is Computationally Efficient



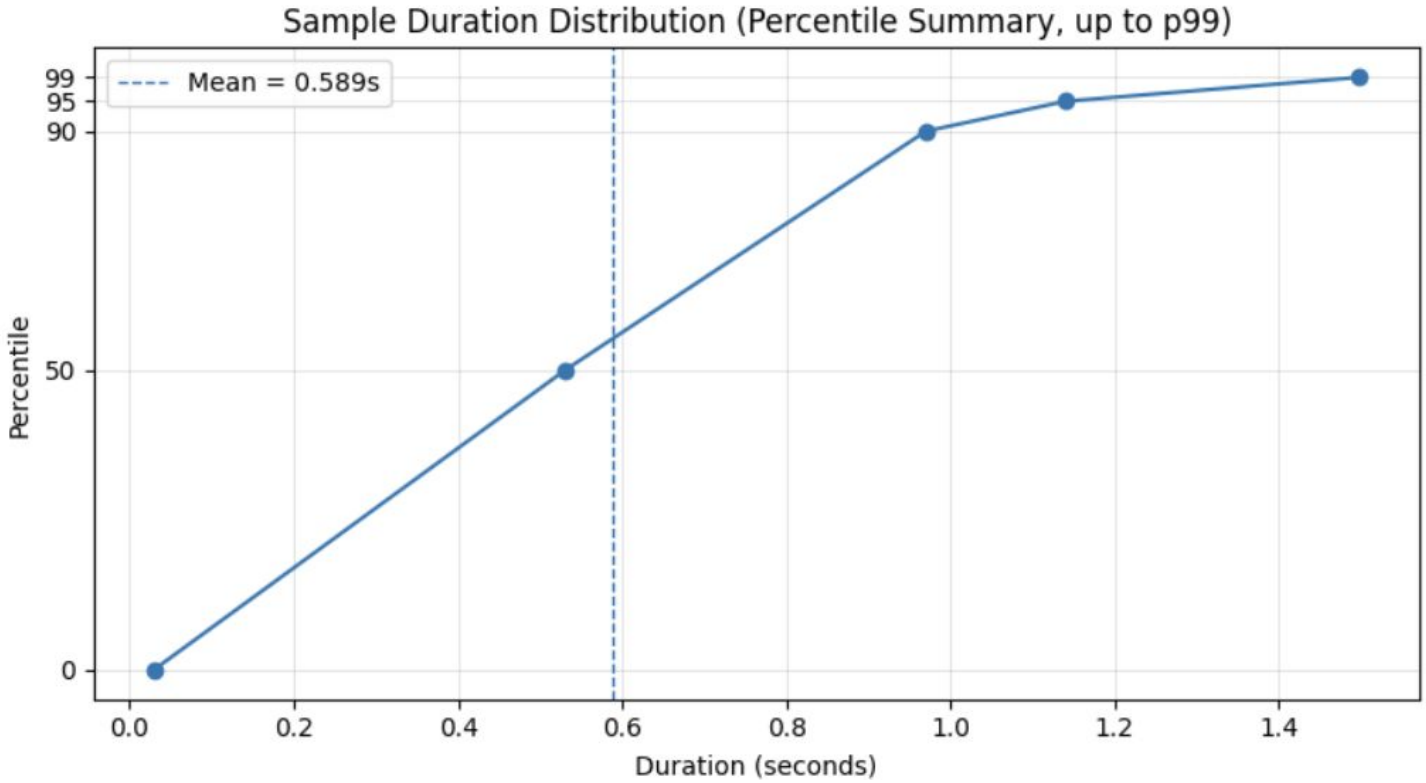
Ablation: Window Size & Stride

Window W (s)	Recall@10 (%) Paper	Tagged
0.96	89.23	71.92
1.92	**92.31**	**77.90**
2.88	**92.31**	74.18
3.84	78.46	68.20

Stride δ	Recall@10 (%) Paper	Tagged
0.96	92.31	77.90
0.48	93.08	**80.76**
0.24	**93.85**	78.77
0.12	**93.85**	77.99

- **Window Size:** 1.92s is the sweet spot. Larger windows add latency without much recall gain. (add distribution of word duration)
- **Stride:** 0.48s is necessary. Larger strides miss terms at boundaries

Gigaspeech Term Distribution

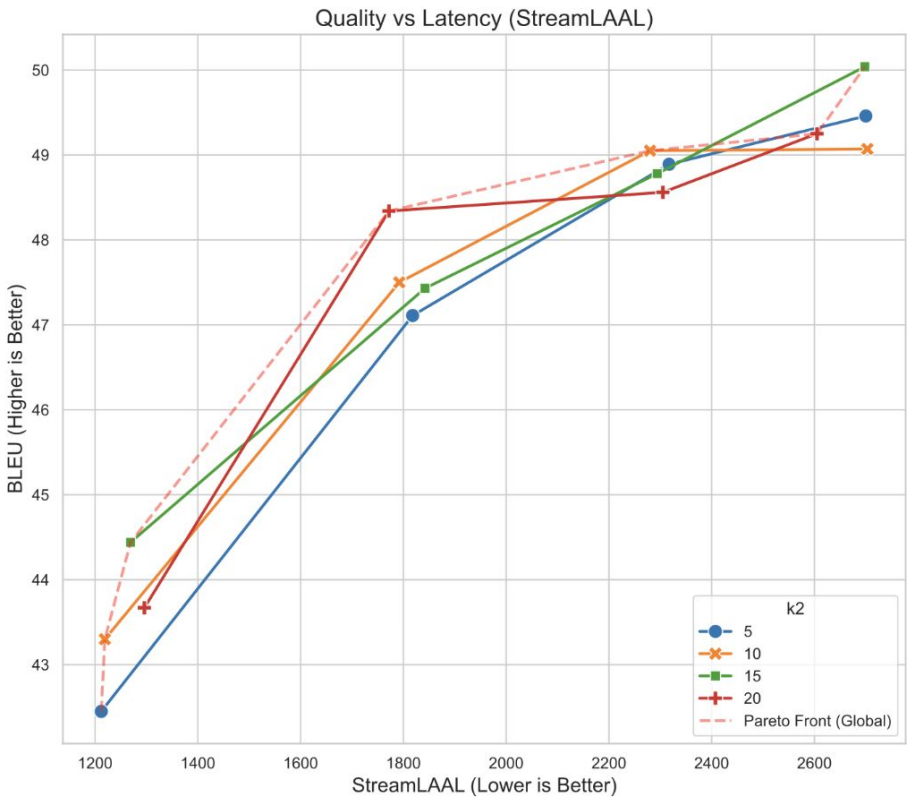


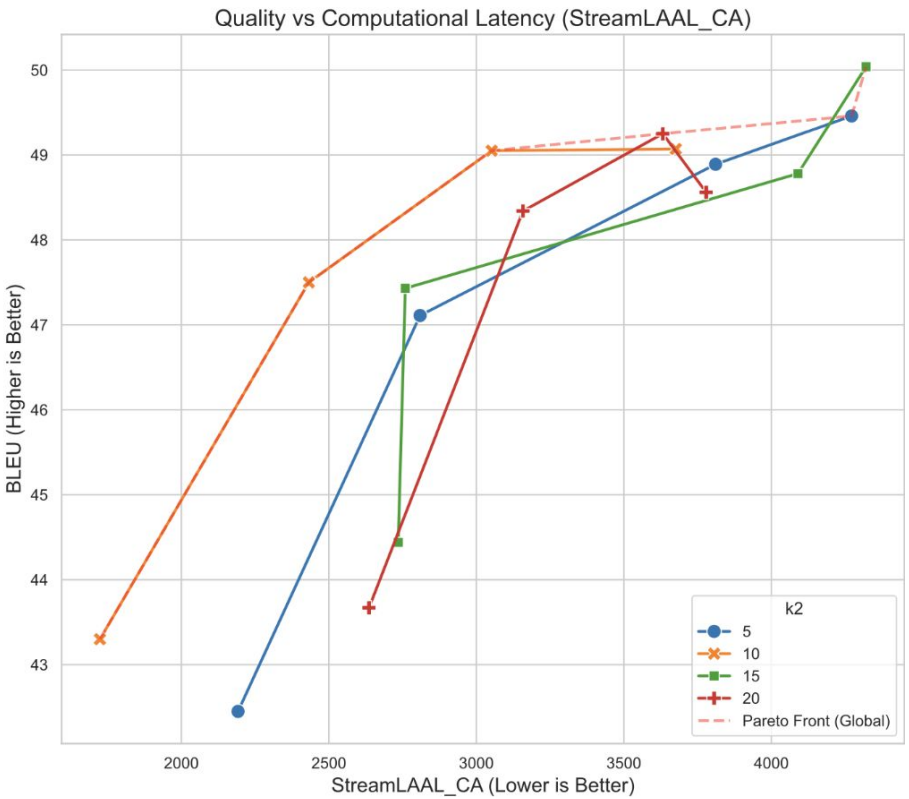
Ablation: Negative Synthesis Strategy

Negative Synth Strategy	BLEU	Terminology Accuracy (%)
Random	28.2 / 37.6	59.4 / 83.5
LLM	47.2 / 47.6	82.3 / 84.5
BGE-M3	47.2 / 48.2	89.6 / 86.0

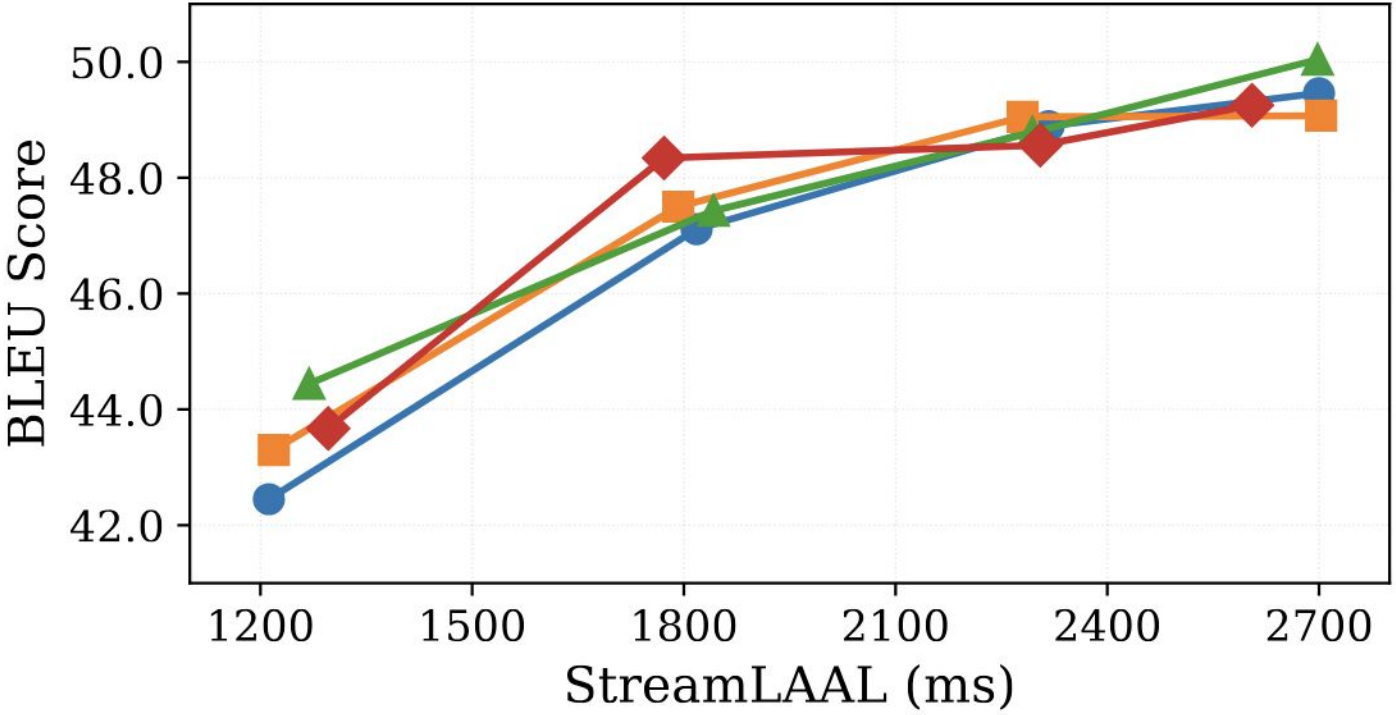
Comparison: Random vs. LLM-generated vs. BGE-M3 (Hard).
Result: BGE-M3 Hard Negatives yield the highest Term Accuracy and BLEU.
Insight: Training with difficult distractors prevents the model from blindly copying retrieved terms

Ablation K2:

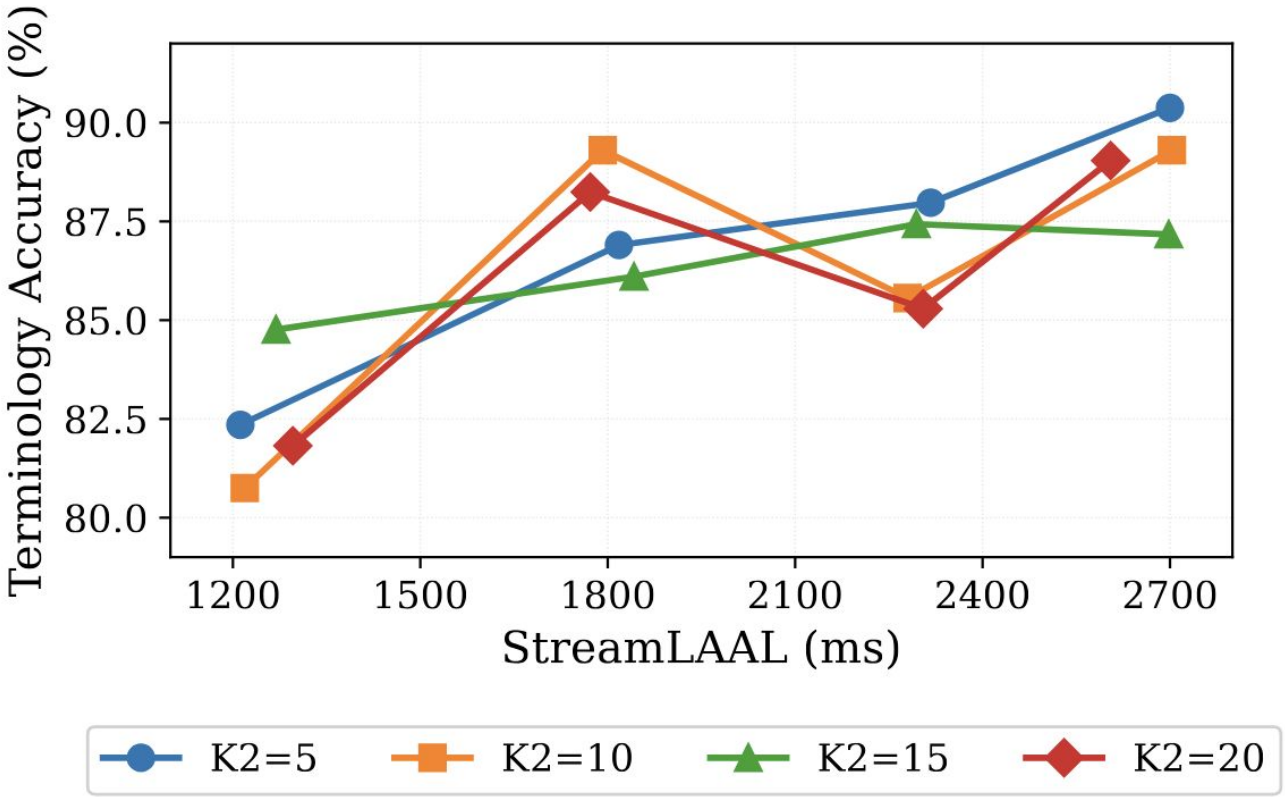

Quality vs Latency (StreamLAAL)
BLEU (Higher is Better)
StreamLAAL (Lower is Better)
50
49
48
47
46
45
44
43
1200
1400
1600
1800
2000
2200
2400
2600
k2
5
10
15
20
Pareto Front (Global)


Quality vs Computational Latency (StreamLAAL_CA)
BLEU (Higher is Better)
StreamLAAL_CA (Lower is Better)
50
49
48
47
46
45
44
43
2000
2500
3000
3500
4000
k2
5
10
15
20
Pareto Front (Global)

Ablation: Top K2 - Quality



Ablation: Top K2 - Term_Acc



Case Study

Performance Overview

- **RASST** achieves **89.3% Term Accuracy** (vs. 73.0% Baseline).
- **BLEU:** 47.50 (RASST) vs. 45.66 (Baseline).

Error Analysis (Three Categories)

- **Cat 1: Success (Retrieval Helps)**
  - Retrieval fixes confusion (e.g., Outputting **CDA** instead of *CDR*).
- **Cat 2: Model Miss (Retrieval Correct, Generation Wrong)**
  - Model ignores the retrieved hint (e.g., Copies **FLR** instead of translating into “最终延迟减少”).
- **Cat 3: Retrieval Miss**
  - Retriever returns irrelevant topics; misses target (e.g., Missing **baseline=**基线).

Observation

- **Over-triggering:** Retrieval occasionally forces synonym changes in non-term segments.

Conclusion

RASST: A specialized framework for streaming speech translation with terminology constraints.
Innovation: Fast sliding-window cross-modal retrieval + Robust training with hard negatives.
Performance: Large gains in term accuracy on ACL 60/60 without sacrificing latency.

Takeaway

- **Database Scalability:** Current experiments use document-level glossaries. Need to test on massive term banks.
- **Dependency:** Performance relies heavily on the quality of the underlying Speech LLM and Retriever.

TODOs

- Enriched Eval (RealSI Domain)
  - Evaluate robustness under larger glossary size
  - Focus metric: term recall / term accuracy under scaled term space

- Enriched Retriever
  - Current offline max recall: 93.4%
  - Hard Negative mining didn’t help (short terms → weak text separation, side effects)
  - Next: TTS-synthesize term audio → audio-modal retrieval to improve recall

Q&A

Ablation thresholding strategy

Thresholding Strategy	Selection Criterion	Recall	F3 Score	Term Acc	Correct / Total
Baseline	Fixed setting	0.93†	–	0.85	318 / 374
Relative Margin	Top-10 relative	–	–	0.75	–
Absolute Threshold	Best F3	–	**0.77**	0.80	300 / 374
Absolute Threshold	Recall ≥ 0.93	**0.935**	–	**0.86**	322 / 374